THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** Student A is correct	72 is the LCM
2	**(d)** 1	They have no common prime factor
3	**(c)** 2	$4 - 6 + 2 = 0$
4	**(a)** $3, -5$	$(x-3)(x+5)$
5	**(c)** Incorrect — the lines intersect	Unique solution
6	**(d)** $(2, 2)$	$x = y = 2$
7	**(c)** Incorrect — it has two equal real roots	$x = -\frac{b}{2a}$ (repeated)
8	**(a)** $x^2 - 8x + 16 = 0$	$(x-4)^2$
9	**(b)** Incorrect — the differences 3, 5, 7 are not equal	Not a constant difference
10	**(d)** 5050	$\frac{100 \times 101}{2}$
11	**(b)** 5 cm	Ratio $\frac{1}{2}$
12	**(a)** AAA	Equiangular triangles
13	**(c)** $\frac{1}{2}$	$\frac{3}{4} - \frac{1}{4}$
14	**(d)** 90°	Theorem 10.1
15	**(b)** 44 cm	$\frac{1}{4} \times 176$
16	**(c)** 14 cm	$r^2 = 196$
17	**(b)** 60π cm²	$l = 10$
18	**(b)** 28	$\frac{24+60}{3}$
19	**(c)** 6	$\frac{30}{5}$
20	**(d)** 0	Impossible

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	(i) $= 3$, rational (ii) $= 5 + 2\sqrt{6}$, irrational	1+1
22	$\alpha + \beta = -\frac{5}{3}$, $\alpha\beta = -\frac{2}{3}$; $\frac{25}{9} + \frac{4}{3} = \frac{37}{9}$	1+1
23	$(3x-2)(2x+1) = 0 \Rightarrow \frac{2}{3}, -\frac{1}{2}$	1+1
24	$(4x-3)(5x-3) = (8x-7)(3x-1) \Rightarrow 2x^2 - x - 1 = 0 \Rightarrow x = 1$ ($x = -\frac{1}{2}$ rejected)	1+1
25	$\left(\frac{18-3}{5}, \frac{16+9}{5}\right) = (3, 5)$	1+1
26	$4 - 4 + \frac{3}{4} + 0 = \frac{3}{4}$	2
27	$20\sqrt{3} \approx 34.64$ m	2
28	$\sqrt{625 - 576} = 7$ cm	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,4), (6,0); (1,0), (0,-1)$. Intersect at $(3,2) \Rightarrow$ consistent	2+1
30	$a = 125, h = 50$; $\Sigma fu = -8 - 8 + 0 + 13 + 12 + 9 = 18 \Rightarrow 125 + \frac{50 \times 18}{50} = 143$	1+1+1
31	$a + 8d = 0$; $a_{29} = a + 28d = 20d$, $a_{19} = a + 18d = 10d \Rightarrow a_{29} = 2a_{19}$	1+1+1
32	$\triangle OPT \cong \triangle OQT$ (RHS) $\Rightarrow \angle POT = \angle QOT$	1+1+1
33	108°; area $= 0.3 \times 154 = 46.2$ cm²	1+2
34	(i) $\frac{45}{50} = \frac{9}{10}$ (ii) $\frac{5}{47}$	2+1
35	$\frac{75}{\sin 60^\circ} = \frac{150}{\sqrt{3}} = 50\sqrt{3} \approx 86.6$ m	1+1+1

	OR: $100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{200}{\sqrt{3}} \approx 115.47$ m	1+1+1
36	$n = 50$; cf 2, 14, 36 ⇒ $20 + \frac{25-14}{22} \times 10 = 25$	1+1+1
	OR: $7 + \frac{40-30}{80-30-16} \times 3 = 7.88$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	$AB = CD = \sqrt{17}$, $BC = DA = \sqrt{2}$ ⇒ parallelogram (3). $AC = \sqrt{13} \neq BD = 5$ ⇒ not a rectangle (2)	3+2
	OR: (i) Mid-point of BC $= (3, 0)$ ⇒ median $= 5$ (ii) $AB^2 = BC^2 = 20$, $AC^2 = 40 = AB^2 + BC^2$ ⇒ right isosceles at B	2+3
38	$\frac{(1+\cos A)^2 + \sin^2 A}{\sin A(1+\cos A)} = \frac{2(1+\cos A)}{\sin A(1+\cos A)}$ (3). $A = 60°$: $\sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}} = 2 \cdot \frac{2}{\sqrt{3}}$ ✓ (2)	3+2
	OR: Expand and use $\sin^2 A + \cos^2 A = 1$ (2); $2 = 1 + 2\sin A \cos A \Rightarrow \sin A \cos A = \frac{1}{2}$, $A = 45°$ (3)	2+3
39	(i) $1540 + 1232 + 718.67 = 3490.67$ cm³ (ii) $440 + 550 + 308 = 1298$ cm²	3+2
	OR: (i) $314 + 261.67 = 575.67$ cm³ (ii) $125.6 + 157 = 282.6$ cm² (iii) ₹565.20	2+2+1
40	$f_1 + f_2 = 18$; $1940 + 30f_1 + 70f_2 = 2880 \Rightarrow 3f_1 + 7f_2 = 94$ ⇒ $f_2 = 10$, $f_1 = 8$	2+2+1
	OR: Mean: $a = 325$, $\Sigma fu = -2$ ⇒ ₹323 (2); median: cf 5, 17 ⇒ $300 + \frac{8}{18} \times 50 \approx 322.22$ (2); mode $= 300 + \frac{6}{14} \times 50 \approx 321.43$ (1)	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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